

Full length article

Anisotropic nanorod-integrated magneto-mechano-electric cascade system for neural electrical stimulation

Dan Wei^a, Xurui Zhao^a, Mingze Zeng^a, Rong Li^a, Wei Zhang^a, Alina Urakova^c, Jie Ding^a, Chengheng Wu^{a,b}, Jing Sun^a, Roman V. Chernozem^c, Hongsong Fan^{a,*}

^a National Engineering Research Center for Biomaterials, College of Biomedical Engineering, Sichuan University, Chengdu 610064, Sichuan, China

^b Institute of Regulatory Science for Medical Devices, Sichuan University, Chengdu 610065, Sichuan, China

^c Piezo- and Magnetoelectric Materials Research & Development Centre, Research School of Chemistry & Applied Biomedical Sciences, National Research Tomsk Polytechnic University, Tomsk 634050, Russia

ARTICLE INFO

Keywords:

Biomaterials
Piezoelectric materials
Anisotropic nanorod
Magneto-mechano-electric cascade
Neural electrical stimulation

ABSTRACT

Piezoelectric materials are now being considered as a potential treatment system for neural electrical stimulation. However, their therapeutic efficacy is limited by insufficient electrical outputs, which are generated by cellular forces and/or tissue motions. Herein, we constructed a high-efficiency magneto-mechano-electric cascade system with magnetic field (MF)-driven wireless electrical stimulation. The anisotropic Fe₃O₄ nanorod (RFO)-integrated poly(vinylidene fluoride-co-trifluoroethylene) copolymer (PVDF-TrFE) fibrous scaffold (PT-RFO) with enhanced piezoelectric β -phase content was prepared via electrospinning technique. Under a remote pulsed MF, the high-efficiency electrical outputs within PT-RFO scaffold could be generated by converting pressure force and deflection force within the contact interface between anisotropic RFO and PVDF-TrFE matrix. As a result, such PT-RFO scaffold could accelerate the repair of injured sciatic nerves and promote the recovery of damaged motor functions in rat models. This work highlights the potential application of a wireless controllable cascade stimulation system integrated multiple designable cues for neural repair and modulation.

Statement of significance: A magneto-mechano-electric cascade scaffold via the integration of anisotropic Fe₃O₄ nanorods and a poly(vinylidene fluoride-co-trifluoroethylene) copolymer (PVDF-TrFE) was developed, with an enhanced piezoelectric phase, coupling area and deflection force. This cascade system can produce high-efficiency electrical signal output, which shows good potential in neural electrical stimulation applications.

1. Introduction

Electrical stimulation, which can effectively regulate the biological behavior of cells through potential changes, has been proven effective in neural electrical stimulation field to promote the repair and regeneration of injured nerves [1,2]. Nowadays, many invasive electrical stimulation systems have been developed. However, these invasive electrical stimulation systems are always accompanied by long-term tissue-traversing and percutaneous wires for connecting bulky electrochemical power sources, which limits the operational convenience and leads an increased infection risk [3]. Besides, most invasive electrical stimulation systems always result in excessive charge accumulation on the cell membrane, which may cause neural stimulus-inertia, consequential inflammation, immune rejection and nerve growth inhibition etc. [4–6]. Thus, developing wireless electrical stimulation systems with

controllable electrical outputs is highly desirable for neurorestoration and neuromodulation.

Piezoelectric materials, such as lead zirconate titanate (PZT), barium titanate (BaTiO₃), poly-L-lactide (PLLA) and polyvinylidene fluoride (PVDF), can convert mechanical forces from different sources into electrical stimulation [7]. Among which, PVDF has received widely attentions due to its outstanding processing feasibility and good sensitivity to mechanical forces [8]. Structurally speaking, PVDF represents five crystalline polymorphs, including α , β , γ , δ , and ϵ phases [9]. Among them, β -phase shows all-trans planar zigzag conformation, which can result in a prominent dipole moment [10]. Hence, increasing the β -phase content of PVDF is the key to enhance its piezoelectric effect for more electrical outputs. Up to now, there are many strategies proposed to solve this issue, including modification with copolymers and combination with electrospinning technology [11,12]. Besides, nanoparticles

* Corresponding author.

E-mail address: hsfan@scu.edu.cn (H. Fan).

<https://doi.org/10.1016/j.actbio.2025.11.007>

Received 24 June 2025; Received in revised form 29 October 2025; Accepted 4 November 2025

Available online 5 November 2025

1742-7061/© 2025 Acta Materialia Inc. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(NPs), acting as the nucleating agents, also have been allowed to incorporate into PVDF to induce β -phase formation [13]. Despite this, using of sole piezoelectric materials even with higher β -phase content *in vivo* is usually driven by cellular force or tissue motion to generate electrical outputs. However, the force amplitude of this source is not large enough, which limits the electrical outputs for effectively regulating the biological behavior of cells.

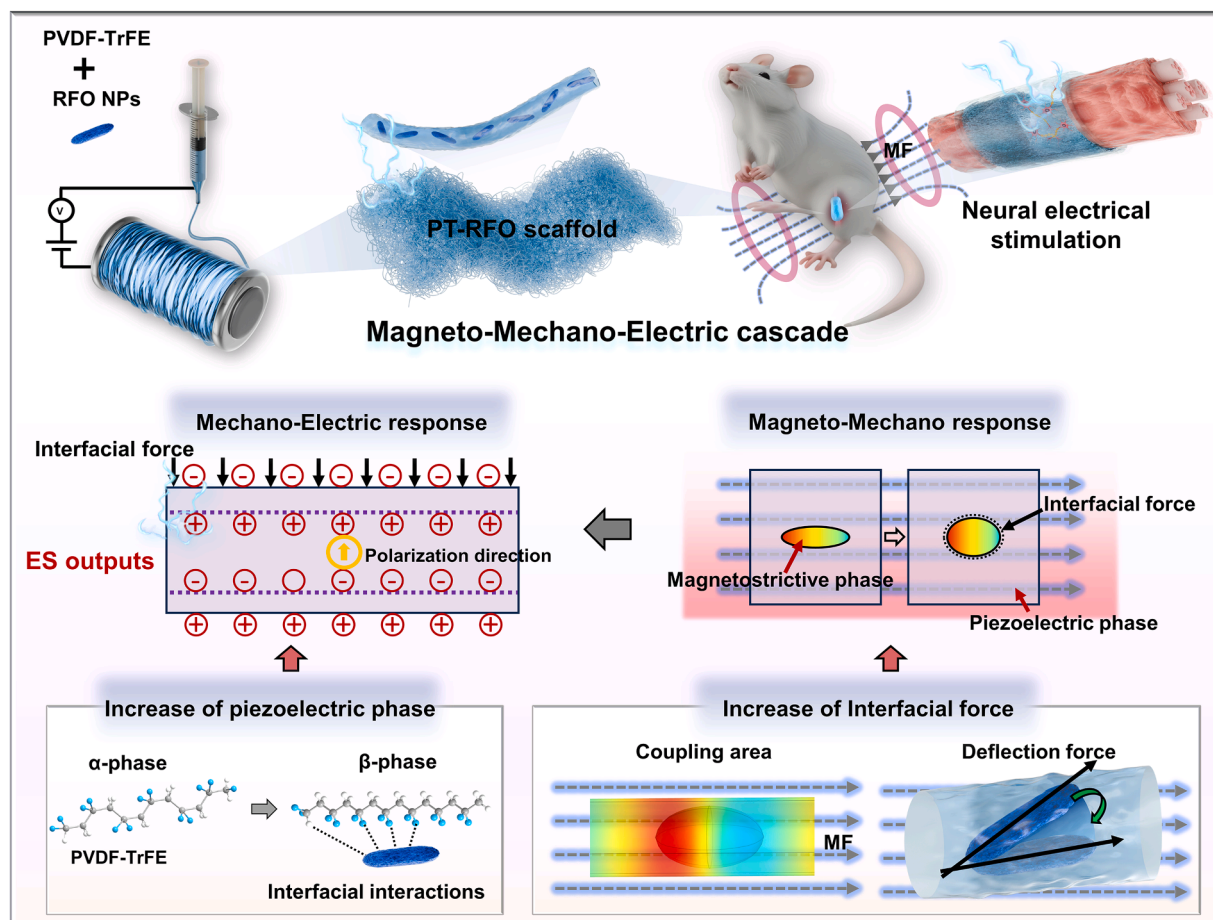
Inspired by the operation rules of a cascade system, integrating wireless, controllable multiple force cues, such as magneto-mechano response, to expand the source of forces, which apply on the piezoelectric materials, may be more helpful to improving the electrical outputs *in vivo* applications [14–17]. By integrating magnetic nanomaterials and piezoelectric materials together, the magneto-mechano response along with the direction of exogenous magnetic field (MF) brings a direct pressure of magnetic nanomaterials applying onto the piezoelectric materials [14]. Then, the piezoelectric materials trigger the alterations in surface potential for generating electrical outputs. A typical magneto-mechano-electric cascade example is magnetoelectric NPs. Choi et.al reported the core-shell structured magnetoelectric NPs by synthesizing Fe_3O_4 NPs onto PVDF NPs [14]. However, without scaffold integration, NPs alone struggle to remain at the target site *in vivo*, and are thus insufficient to induce tissue regeneration. The integration of magnetic nanomaterials into piezoelectric scaffolds represents an alternative and promising approach. Some studies introducing magnetic NPs (e.g. Fe_3O_4 , CoFe_2O_4 and $\text{CoFe}_2\text{O}_4/\text{BaTiO}_3$ NPs) into piezoelectric materials, and the electrical outputs of these scaffolds can be regulated by the amount of embedded magnetic NPs [15,18,19]. However, the output voltage of them still needs to be further optimized, especially in the case of low-intensity MF for biomedical applications.

Here, we present a magneto-mechano-electric cascade system for outputting electrical signals efficiently, which can promote peripheral nerve injury (PNI) regeneration and repair (Scheme 1). By employing electrospinning, the anisotropic Fe_3O_4 NPs-integrated poly(vinylidene fluoride-co-trifluoroethylene) copolymer (PVDF-TrFE) fibrous scaffold (PT-RFO) with high β -phase content is obtained. The introduction of anisotropic Fe_3O_4 (RFO) NPs into PVDF-TrFE promote the formation of electroactive β -phase via interfacial interactions. Compared to isotropic Fe_3O_4 (SFO) NPs, the use of RFO NPs increases the coupling area between magnetostrictive Fe_3O_4 phase and piezoelectric PVDF-TrFE phase. Besides, under an external pulsed MF, the initial random distributed RFO NPs are prone to micro-vibration along with the direction of applied MF, resulting in an additional continuous deflection force onto the contact interface between RFO NPs and PVDF-TrFE matrix. Benefiting from the high-efficiency magneto-mechano-electric cascade conversion, PT-RFO scaffold with high electrical outputs show good potential in the promotion of PNI regeneration, demonstrated by the repair of rat's sciatic nerve injury. More importantly, this study provides a new strategy which is applicable to neurorestoration and neuromodulation via integrating cascade responsive material cues.

2. Experiment

2.1. Reagents

$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ was purchased from Aladdin Reagent Co. Ltd. (Shanghai, China). Trisodium citrate dehydrate (Na_3CT), ammonium acetate (NH_4OAc), ethylene glycol (EG), Diethylene Glycol (DEG), N, N-Dimethylformamide (DMF), Acetone (AC) and aqueous ammonia (30 %)



Scheme 1. Schematic of magneto-mechano-electric cascade system with enhanced electrical outputs for the promotion of neurorestoration and neuromodulation.

were all obtained from Kelong Reagent Co. Ltd. (Chengdu, China). Polyacrylic acid (PAA) was purchased from Macklin Reagent Co. Ltd. (Shanghai, China). PVDF-TrFE was purchased from Solvay (France). All primary antibodies for immunofluorescence staining were obtained from Abcam (USA) and all fluorescently labeled secondary antibodies were purchased from Invitrogen (USA). Unless otherwise specified, all other chemicals were acquired from Chengdu Kelong Chemicals.

2.2. Fabrication of SFO and RFO NPs

The SFO NPs were synthesized via solvothermal reaction. In brief, the reagents of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ (1.35 g), Na_3CT (0.46 g), NH_4OAc (3.854 g) and EG (70 mL) were added into a beaker. The mixture was maintained vigorously stirring for 1.5 h at room temperature. The black mixture was transferred into a Teflon-lined stainless-steel autoclave of 100 mL. After reaction for 8 h at 200 °C, the mixture was cooled to room temperature. Then, the SFO NPs were collected using a magnet, followed by alternate washing with ethanol and deionized water (DIW) three times. Finally, the SFO NPs were dispersed in ethanol (20 mL) for further use.

The RFO NPs were synthesized via a two-step reaction. First, FeOOH NPs were prepared via hydrothermal reaction. FeCl_3 (1.0812 g) was dissolved in DIW (40 mL). Then, the solution was transferred into a three neck round bottom flask, followed by heating at 87 °C for 18 h. The product was washed by DIW for three times, and then collected by centrifugation. After that, the collected sediment was dispersed in DIW (15.4 mL), followed by adding 1 mL of 80 mg/mL PAA solution. Then the FeOOH -PAA NPs were collected and resuspended in DEG solution (28 mL). Then, the mixture was reacted for 8 h at 220 °C under nitrogen protection to obtain RFO NPs. Finally, the resulting RFO NPs were collected using a magnet after washing, and dispersed in ethanol (20 mL) for further use.

2.3. Characterizations of SFO and RFO NPs

The morphology of SFO and RFO NPs were characterized using a field emission scanning electron microscopy (SEM, S-4800, Hitachi, Japan). The structure and energy-dispersive spectroscopy (EDS) mappings were characterized using transmission electron microscopy (TEM, JEM-2100Plus, Japan). The size-distribution of SFO and RFO NPs were calculated with SEM images by ImageJ software.

For the calculation of the SSA of a single SFO and RFO NP, following formulas were used. A SFO NP is considered as a sphere, and the SSA of it was defined as (1):

$$SSA_{(SFO)} = \frac{3}{R} \quad (1)$$

where R is radius of the sphere.

A RFO NP is considered as a spheroid, and the SSA of it was defined as (2):

$$SSA_{(RFO)} \approx 4\pi \left[\frac{(a^p b^p + a^p c^p + b^p c^p)}{3} \right]^{1/p} \quad (2)$$

where a, b, and c are the three semi-axes of the ellipsoid, with $a \geq b \geq c$. $p \approx 1.6075$

X-ray photoelectron spectroscopy (XPS) data were acquired using an X-ray diffractometer (AXIS Ultra DLD, Co $K\alpha = 1.7890 \text{ \AA}$). The magnetic properties were tested with a Vibration Sample Magnetometer (VSM, model BHV-525, Riken Japanese Electronics Company). X-ray diffraction (XRD) data were acquired by an XRD (X'Pert Pro MPD, Philips, Netherlands). Then, XRD patterns were analyzed by the Rietveld refinement using in the Match software and COD database. Microstress of the structure of the formed NPs was defined as (3):

$$\frac{\Delta d}{d} = \frac{\beta}{4 \tan \theta_{hkl}} \quad (3)$$

where β is peak width at half height, and θ_{hkl} is the diffraction angle.

2.4. Fabrication of PVDF-TrFE (PT), SFO-integrated PVDF-TrFE (PT-SFO) and PT-RFO scaffolds

PT scaffold was prepared by electrospinning the solution of PVDF-TrFE using #27 G nozzle with a flow speed of 1 mL h^{-1} . In short, the electrospinning PVDF-TrFE solution was prepared by dissolving PVDF-TrFE powder in DMF/acetone solution (V: V = 3:2) under sonication for 30 mins, to obtain a concentration of 16 wt %. The electrospinning process was performed with a high direct current voltage of $-2 \sim 13 \text{ kV cm}^{-1}$ at room temperature and 35 % humidity. PT-SFO and PT-RFO scaffolds were prepared by electrospinning the PVDF-TrFE (5 mL) with SFO (20 mg) and RFO NPs (20 mg), respectively. As-received PT, PT-SFO and PT-RFO scaffolds were stored in vacuum for further use, respectively. To obtain collagen-coated PT, PT-SFO and PT-RFO scaffolds, collagen solution (0.5 mg/mL) was electrosprayed onto these scaffolds with a flow speed of 3 mL h^{-1} and a high direct current voltage of $-2 \sim 10 \text{ kV cm}^{-1}$.

2.5. Characterizations of PT, PT-SFO and PT-RFO scaffolds

The morphology and structural distribution were investigated with SEM and TEM. The size of fibers was calculated by ImageJ Software. Fourier transform infrared spectrometer (FT-IR, IRTTracer100, Japan, SHIMADZU) was applied to verify the formation of β -phase. The fraction of the β -phase, $F(\beta)$, in the PVDF-based scaffolds can be calculated from the FT-IR spectra using the Lambert-Beer law as (4) [20]:

$$F\beta = \frac{A_\beta}{\left(\frac{K_\beta}{K_\alpha} \right) A_\alpha + A_\beta} \quad (4)$$

where A_α and A_β are the absorbance at 764 cm^{-1} and 840 cm^{-1} , respectively, and K_α ($6.1 \times 10^4 \text{ cm}^2 \text{ mol}^{-1}$) and K_β ($7.7 \times 10^4 \text{ cm}^2 \text{ mol}^{-1}$) are the absorption coefficients at the respective wave numbers.

An XRD (X'Pert Pro MPD, Philips, Netherlands) was used to measure the crystalline phases of the scaffolds. X-ray photoelectron spectroscopy (XPS) data were acquired using an X-ray diffractometer (AXIS Ultra DLD, Co $K\alpha = 1.7890 \text{ \AA}$). For piezoelectricity test, atomic force microscopy equipped (AFM, SCM-PIT, Bruker) with conductive platinum-iridium silicon cantilevers was used. The piezoelectric coefficients ($|d_{33}|$) of these scaffolds were examined with a meter (YE2730A d33 meter, Sinocera, China). The electrical output voltages of the three scaffolds under different MF intensities were recorded using two methods. Based on previous studies [21,22], we prepared a test device by encapsulating a scaffold film ($1.0 \times 1.0 \text{ cm}^2$) with a polyimide (PI) membrane, aluminum (Al) foil, and copper (Cu) wires. The test device was then placed in a platform consisting of several glass slides, with the magnet positioned on top (Figure S1). By periodically loading and removing the magnet, we recorded the measured voltage values. The magnitude of the MF can be altered by changing the number of magnets used. To test the output voltages of the three scaffolds under the pulsed magnetic field applied by a Helmholtz coil, we encapsulated the scaffold film with Cu foil and positioned it at the center of the coil. A test setup was then created, consisting of a pulsed direct current power supply, a Helmholtz coil, and an oscilloscope (Figure S2). The magnitude of the MF can also be adjusted by changing the power output of the power supply. To evaluate *in vitro* degradation, PT, PT-SFO, and PT-RFO specimens ($n = 4$, $10 \times 10 \text{ mm}$) were immersed in 10 mL PBS (pH 7.4) and incubated at 37 °C for up to 28 days. At 0, 14, 21, and 28 days, samples were collected, photographed, rinsed, lyophilized, gold-sputtered, and analyzed by SEM (ZEISS Sigma 360, Germany).

2.6. COMSOL simulation

The differences in magnet-driven piezoelectric properties were simulated using COMSOL Multiphysics software. The MF was built by a rectangular coil, which producing a MF intensity ranged from 10 to 200 mT. For PT-SFO model, the PVDF-TrFE fiber was constructed as a cylinder (diameter:250 nm) with encapsulated SFO spheres (diameter: 200 nm). For PT-RFO model, the encapsulated RFO NPs were defined as ellipsoids (long axis: 380 nm, short axis:200 nm). The physical parameters of the materials were taken from the library of COMSOL.

2.7. In vitro biocompatibility test

PT, PT-SFO and PT-RFO scaffolds were cut into square samples with a size of 1 cm and sterilized with UV irradiation for 1 h. Rat adrenal pheochromocytoma cells (PC12) cells were cultured in Dulbecco's modified Eagle's medium (DMEM, Servicebio, China) supplemented with 10 % fetal bovine serum (Zhejiang Tianhang, China) and 1 % penicillin-streptomycin solution (PS; HyClone, USA). Subsequently, PC12 cells were seeded on PT, PT-SFO and PT-RFO scaffolds with a concentration of 4×10^4 cells/mL, respectively. For MF loading, the cells were stimulated by a Helmholtz coil (15 mT, duty of 80 %, 100 Hz, the MF waveform is rectangular waveform with rounded corners) for 15 min every day. The Helmholtz coil system (200 mm inner diameter, 100 turns) was powered by a programmable pulsed direct current supply (0–30V/0–5A). The field intensity was precisely calibrated using a gaussmeter (TM-801EXP, KANETEC, Japan). Live (fluorescein diacetate, FDA)/Dead (propidium, PI) staining was adopted to assess cell viability. For immunofluorescence staining, NF, Tuj1 and PSD95 were performed followed by a procedure according to the user guide. The fluorescence intensity was calculated with ImageJ Software.

2.8. Animals

All animal experiments were carried out according to the ARRIVE (Animal Research: Reporting of *In Vivo* Experiments) guidelines. The experimental procedures were reviewed and approved by the Animal Care and Use Committee of Sichuan University (approve number: SYXK (Sichuan): KS2022863). Female Sprague-Dawley (SD) rats (200–220 g, Chengdu Dossy) were randomly divided into five groups ($n = 5$ /group): sham, TBI, PT, PT-RFO (-MF) and PT-RFO (+MF) groups. After anesthesia by intraperitoneally injecting pentobarbital sodium (50 mg kg⁻¹), the right sciatic nerve was exposed from a transgluteal access. Then, a crush injury model of rats was created by using vascular clips to the sciatic nerve (5 mm) for a duration of 5 min. For sham group, the sciatic nerve was exposed and then sutured without any operations. For PNI group, the sciatic nerve was crushed without any implanted materials. For PT group, the PT scaffold was wrapped onto the sciatic nerve. For PT-RFO (-MF) and PT-RFO (+MF) groups, the PT-RFO scaffold was wrapped onto the sciatic nerve, respectively. For MF loading, the rats were restrained in a cylindrical restrainer, which was positioned inside a Helmholtz coil (15 mT, duty of 80 %, 100 Hz, the MF waveform is rectangular waveform with rounded corners) with the restrainer's upper end fixed 15 cm below the coil's top for 15 min every day (Figure S3). The field intensity was precisely calibrated using a gaussmeter (TM-801EXP, KANETEC, Japan).

2.9. Walking track analysis

At 28 days, walking track analysis was used pre-operatively to evaluate the nerve functional recovery. The rats were allowed to traverse a 1 m runway, and their footprints were recorded. Three parameters, including print length (PL, distance from heel to toe3), toe spread (TS, distance from toe1 to toe5) and intermediate toe spread (IT, distance from toe2 to toe4) are defined. The prefixes R and L denote measurements from the right (injured) and left (normal) paws, respec-

tively. The sciatic functional index (SFI) was calculated as (5):

$$SFI = -38.3 \left(\frac{RPL - LPL}{LPL} \right) + 109.5 \left(\frac{RTS - LTS}{LTS} \right) + 13.3 \left(\frac{RIT - LIT}{LIT} \right) - 8.8 \quad (5)$$

2.10. Histological and immunofluorescence staining

At the scheduled time point, SD rats were anesthetized with an intraperitoneal injection of sodium pentobarbital (100 mg kg⁻¹). The sciatic nerves of rats were isolated, paraffin-embedded and cut into 5 μm thick slices. Hematoxylin and eosin (HE) staining and immunofluorescence staining (IL-10, TNF-α, NF, Tuj1 and S100) were performed, followed by a procedure according to the user guide. The fluorescence intensity was calculated with ImageJ Software.

2.11. Statistical analysis

All the data are showed as mean ± standard deviation (SD). One-way ANOVA and Two-way ANOVA were performed using GraphPad Prism6 software. Significant differences were considered at the value of *p*-value of **p* < 0.05, ***p* < 0.01, or ****p* < 0.001.

3. Results and discussion

3.1. Preparation of magnetic Fe₃O₄ NPs with different morphology

In this work, we firstly synthesized two types of NPs: SFO and RFO NPs (Fig. 1A). SFO NPs were prepared via a simple one-step solvothermal reaction. To obtain RFO NPs, we employed a surface protection approach by transforming non-magnetic FeOOH NPs with PAA treatment into magnetic RFO NPs. Fig. 1B (i) shows SEM images of SFO and RFO NPs, suggesting their entirely different morphology. The TEM images and EDX mapping analysis were also acquired as shown in Fig. 1B (ii and iii), showing their successful synthesis. Then, EDS patterns showed the elements of Fe and O presented in both SFO and RFO NPs and the Fe:O ratios of them are about 3:4, proving the crystalline phase of Fe₃O₄ (Fig. 1C). By statistical calculation, the average diameter of SFO NPs was around 210 nm (Figure S4), while the average length of RFO NPs was around 379 nm and average diameter was approximately 82 nm (Fig. 1D). As shown in Fig. 1E, the Zeta potential of SFO NPs was −215.833 mV and that of RFO NPs was −817.67 mV. We also calculated the SSA, showing the average SSA_{RFO} was ~ 0.0538 nm⁻¹, higher than that of SFO NPs (~ 0.0143 nm⁻¹) (Fig. 1F).

XPS confirmed the presence of C 1 s, O 1 s and Fe 2p peaks in two types of Fe₃O₄ NPs (Fig. 1G). The fitting of Fe 2p region of both SFO and RFO NPs confirmed the formation of the magnetite structure with three typical peaks corresponding to Fe_{Oh}²⁺ (~710.1 eV), Fe_{Oh}³⁺ (~711.2 eV), and Fe_{Th}³⁺ (~713.1 eV), indicating the formation of cubic inverse spinel structure [23]. C 1 s region of both SFO and RFO NPs was fitted with three peaks associated to C—C/C—H (285 eV), C—O (~286.4 eV) and C = O (~288.6 eV) functional groups, which can be related to the carbon-containing reagents used during synthesis process, such as EG, DEG, and PAA. It's worth noting that, the fitting of O1s region revealed the increased content of C—O and C = O functional groups in RFO NPs, which can be contributed to the carboxyls from PAA [24,25]. Subsequently, XRD patterns showed the characteristic peaks corresponding to Fe₃O₄ (magnetite, #COD: 96–900–9769) were both observed in SFO and RFO NPs (Fig. 1H). By further analyzing by the Rietveld refinement method, the phase ratio results showed there were no impurities in SFO and RFO NPs (Figure S5, S6 and Table S1). Moreover, the microstress in RFO NPs was lower than that in SFO NPs, indicating less lattice distortion [26]. Then, the magnetic property was investigated by a VSM. As VSM curves in Fig. 1I shown, despite multistep reactions, RFO NPs still

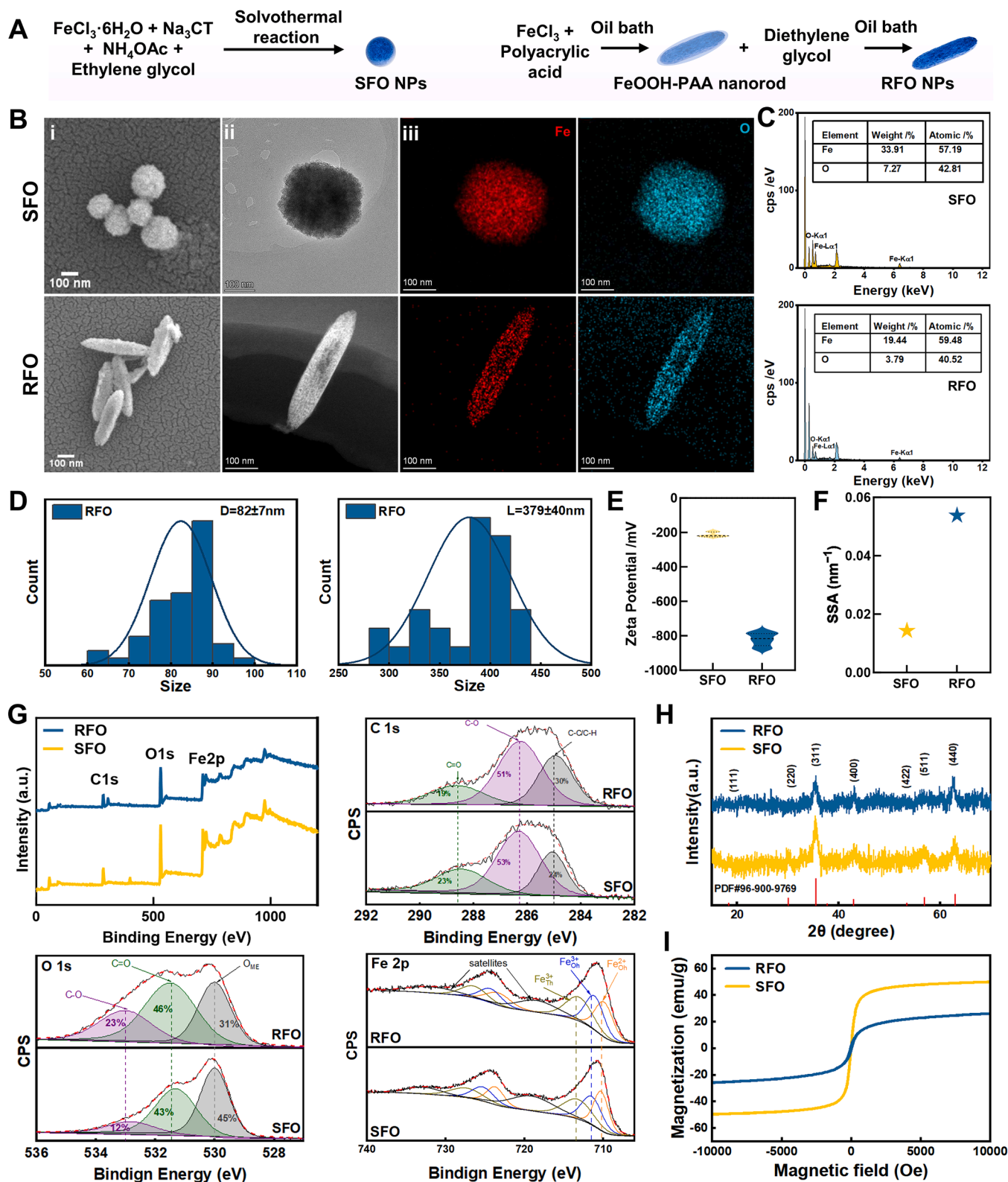


Fig. 1. The synthesis and characterization of SFO and RFO NPs. (A) Schematic of synthetic routes of SFO and RFO NPs. (B) (i) SEM images, (ii) TEM images and (iii) EDX mapping images of SFO and RFO NPs. (C) EDS analysis of SFO and RFO NPs. (D) Statistical length (L) and diameter (D) of RFO NPs with ImageJ software. (E) Zeta potential of SFO and RFO NPs. (F) Calculated SSA results of SFO and RFO NPs. (G) XPS spectra of C 1s, O 1s and Fe 2p region of SFO and RFO NPs. (H) XRD patterns of SFO and RFO NPs. (I) Hysteresis loops of SFO and RFO NPs.

possessed a saturation magnetization of 26 emu g^{-1} , indicating the good magnetic responsiveness.

3.2. Construction of magneto-mechano-electric cascade scaffolds

Considering co-polymerizing PVDF-TrFE possesses higher β -phase crystallization, compared to pure PVDF, which can produce more dipole moments for the improvement in piezoelectric responsiveness [27]. The

magneto-mechano-electric cascade scaffolds were fabricated by integrating SFO or RFO NPs into piezoelectric PVDF-TrFE fibrous matrix via electrospinning technique to yield out PT-SFO and PT-RFO scaffolds (Fig. 2A). Upon electrospinning, a high voltage between the spinning fluid and the collector creates an electric field that elongates the fluid into a Taylor cone. Once the voltage exceeds a threshold, a charged jet breaks free and is drawn to the collector. During this process, the uniaxial stretching is applied to orient the PVDF chains dipoles, allowing

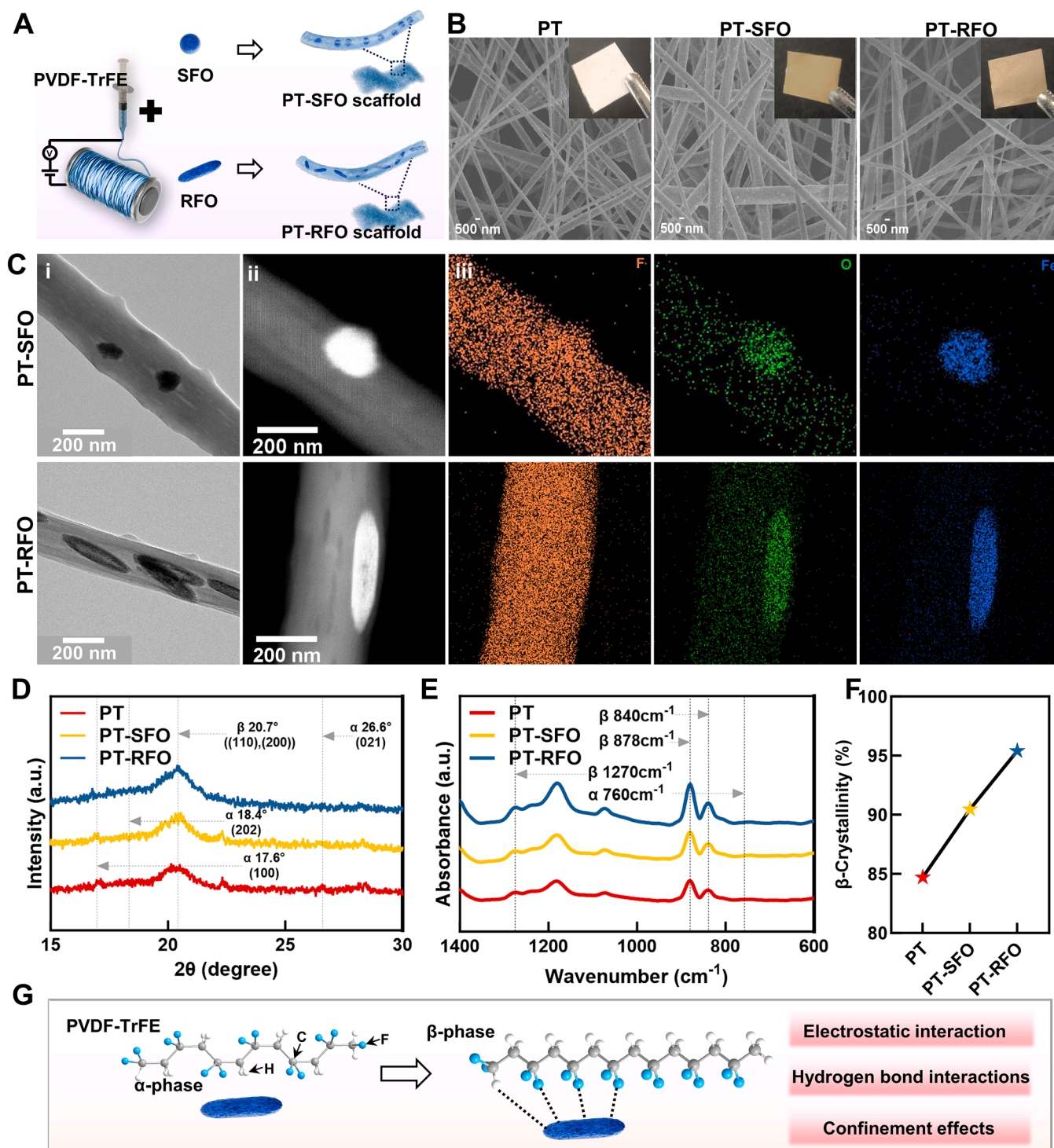


Fig. 2. Fabrication and characterization of magneto-mechano-electric cascade scaffolds. (A) Schematic of preparation process of PT, PT-SFO and PT-RFO scaffolds. (B) SEM images of PT, PT-SFO and PT-RFO scaffolds. (C) (i) SEM images, (ii) TEM images and (iii) EDX mapping images of PT-SFO and PT-RFO scaffolds. (D) XRD patterns. (E) FT-IR spectrum. (F) Calculated β -phase contents. (G) The schematic of PT-RFO scaffold showing more β -phase content due to the enhanced interfacial interactions.

the α -to- β crystalline phase changes [28]. In addition, the negative $-\text{CF}_2$ dipoles within PVDF chains can be oriented along an applied voltage direction effectively, that is in-situ polarization [29]. As Fig. 2B, C, S7 & S8 shown, both PT-SFO and PT-RFO scaffolds showed fibrous structure and successful magnetic NPs embedding. Under an external electric field, dispersed SFO and RFO NPs in initial PVDF-TrFE solution would be covered by PVDF-TrFE layer respectively instead of locating on the surface of PVDF-TrFE layer, due to the surface tension forces [30]. Degradation tests revealed that all three scaffolds underwent time-dependent degradation in PBS, characterized by increasingly loosened microstructures (Figure S9).

Then, we investigated the β -phase crystallization in the magneto-mechano-electric cascade scaffolds with different magnetic Fe_3O_4 NPs. Fig. 2D depicts the XRD patterns of the three scaffolds. The PT scaffold was dominated by the nonpolar α -phase with characteristic peaks at $2\theta = 17.6^\circ$ (corresponding to (100) plane), 18.4° (corresponding to (202) plane) and 26.6° (corresponding to (0 2 1) plane), and a low-intensity β -phase characteristic peak at $2\theta = 20.7^\circ$ (corresponding to (1 1 0), (2 0 0) plane). The respective addition of SFO and RFO NPs into PVDF-TrFE both led to decreased α -phase content, and increased β -phase within in PT-SFO and PT-RFO scaffolds, respectively. In detail, the (1 1 0), (2 0 0) planes of β -phase were found developed within PT-SFO and PT-RFO scaffolds.

The crystallinity of three scaffolds was further analyzed by FT-IR spectra (Fig. 2E). The vibrational peaks at 840 cm^{-1} , 878 cm^{-1} and 1270 cm^{-1} correspond to the β -phase. Obviously, the intensities of these peaks in PT-RFO scaffolds were higher than PT-SFO scaffolds. Then, the β -phase crystallinity content was calculated from FT-IR spectra (Fig. 2F). The results showed that the content of β -phase in PT-SFO scaffold was 90.46 %, and 95.39 % in PT-RFO scaffold, both were higher than that in pure PT scaffold (84.69 %). Apparently, PT-RFO scaffold with anisotropic RFO NPs possessed the highest β -phase content, which is beneficial to the electrical outputs under an external force of the same magnitude.

Fe_3O_4 NPs, as nanofillers, have been proven to promote β -phase formation within PVDF-based materials. This process can be regulated by electrostatic interactions between the negatively charged surface of Fe_3O_4 NPs and the positively charged $-\text{CH}_2$ groups of PVDF chains, facilitating the formation of anion- CH_2 dipoles [31–34]. As demonstrated in Fig. 1E, the SFO and RFO NPs are negatively charged on their surfaces. Hence, compared to pure PT scaffolds, the PT-SFO and PT-RFO scaffolds showed higher β -phase content. In addition, due to more C—O and C = O functional groups provided by PAA protection method, demonstrated in Fig. 1G, the surface charge of RFO NPs is higher than that of SFO NPs. Hence, stronger electrostatic interactions between RFO NPs and PVDF chains is formed, leading to the highest β -phase content within PT-RFO scaffold. On the other hand, according to the FT-IR spectra of $-\text{OH}$ groups ($\sim 3750\text{--}2500\text{ cm}^{-1}$) shown in Figure S10, the features at 3414 and 3338 cm^{-1} within PT-SFO and PT-RFO scaffolds can be considered as the red-shift of $-\text{OH}$ groups due to hydrogen bond formation, which is also favorable for the formation of β -phase [33–35]. Compared to isotropic nanostructures, the anisotropic nanostructures, including nanorods or nanoplanars can promote the formation of β -phase by positioning binding sites through confinement effects [33, 36]. Besides, the larger SSA of RFO NPs can also contribute to these interactions, because of the larger coupling area between magnetostrictive Fe_3O_4 phase and piezoelectric PVDF-TrFE phase. Overall, the introduction of RFO NPs into PVDF-TrFE derived the ability to form more electroactive β -phase via electrostatic interactions, hydrogen bond interactions and confinement effects (Fig. 2G).

3.3. Magnetic-controlled electrical outputs in magneto-mechano-electric cascade scaffolds

The β -phase content of PVDF-based materials is high related to the piezoelectricity. Hence, the piezoelectric performance of PT, PT-SFO,

PT-RFO scaffolds were investigated by piezoelectric force microscopy (PFM). Fig. 3A shows the local single-point phase recordings. It clearly shows the ferroelectric phase response of the three scaffolds. When a positive direct current bias voltage increasing, the dipoles of PVDF crystalline domains within the three scaffolds begin to rearrange, showing related phase change response [37]. It is noted that, the application of external MF hadn't altered the ferroelectric responsiveness of PT-SFO and PT-RFO scaffolds, showing almost coincident hysteresis loops with or without MF, respectively.

The amplitude hysteresis response (butterfly loop) was also measured (Fig. 3B). Without MF, the coercive voltages of PT-RFO scaffold measured by the hysteresis curves were -0.58 V and 0.62 V , which was lower than that of pure PT (-0.82 V and 1.66 V) and PT-SFO scaffolds (-0.63 V and 1.89 V). A reduction in coercive voltage of PT-RFO scaffold means the relatively low operating voltage for polarization switching, which is benefit for low-cost practical applications. When MF (100 mT) was applied, the coercive voltages of PT-RFO scaffold changed to -0.47 V and 1.19 V , and this variation in coercive voltage indicates the mechanical forces generated in the magnetostrictive RFO NPs can be converted to piezoelectric PVDF-TrFE matrix, which promotes the polarization reversal process in PT-RFO scaffold. Moreover, for PT-RFO scaffold under MF, the positive coercive voltage vibration ($\delta = 0.57\text{ V}$) became higher than the negative coercive voltage vibration ($\delta = 0.11\text{ V}$). The asymmetric increment dedicates that the center of piezoresponse loop is offset, which is assumed to be caused by the formation of electric field because of the magnetoelectric effect [38]. The PT-RFO scaffold represented a clearly higher amplitude compared to other scaffolds with or without external MF, which was in accordance with the trend of their β -phase contents. Then, the piezoelectric coefficients were calculated from PFM results (Fig. 3C). The PT-RFO scaffold showed the highest $|d_{33}|$ of 42.67 pm/V , which was almost 1.23 times higher than that of PT-SFO and 2.1 times higher than that of pure PT scaffold. Such a high $|d_{33}|$ value is also higher than other reported PVDF-based scaffolds, films and hydrogels (Table S2) [39–44].

To measure the electrical outputs of PT, PT-SFO, PT-RFO scaffolds with or without external MF, we fabricated a flexible test device composed of a scaffold film, PI membrane, Al foil and Cu wires (Fig. 3D). Then, different magnets were periodically placed on and removed from the top, catching the voltage value corresponding to the moment when the magnet is initially placed, respectively. Despite the increase in MF intensity, pure PT scaffold showed no peak-to-peak values of output voltage, as there is no external force to trigger the piezoelectric response (Fig. 3E). In contrast to that, the values of output voltage within both PT-SFO and PT-RFO scaffolds increased as the increasing of applied MF intensity, as the magnetic NPs undergo rapid shape deformation and deflection. This deformation generates a localized mechanical force that is directly transmitted to the surrounding PVDF-TrFE matrix, thereby eliciting a piezoelectric response. It's worth noting that the PT-RFO scaffold showed higher output voltage than that of PT-SFO scaffold under the same MF intensity. Taking a low-intensity MF of 15 mT as an example, the output voltage of PT-RFO scaffold was $\sim 65\text{ mV}$, higher than that of PT-SFO scaffold ($\sim 35\text{ mV}$). We also measured the output voltages of these three scaffolds under a pulsed MF with different intensities using a Helmholtz coil connected to an oscilloscope. The results showed that the PT-SFO scaffold had the highest voltage value (Figure S11). The possible reasons should be attributed to that PT-RFO scaffold possessed higher piezoelectric β -phase content and more efficient mechano-electric conversion. In addition, the amplitude of output voltage generated by the magneto-mechano-electric cascade PT-RFO scaffold can be easily controlled by the applied MF intensity.

3.4. Magnetic-triggered mechano-electric conversion in magneto-mechano-electric cascade scaffolds

Previous results had shown that PT-RFO scaffold possessed higher piezoelectric β -phase content, which should trigger higher output

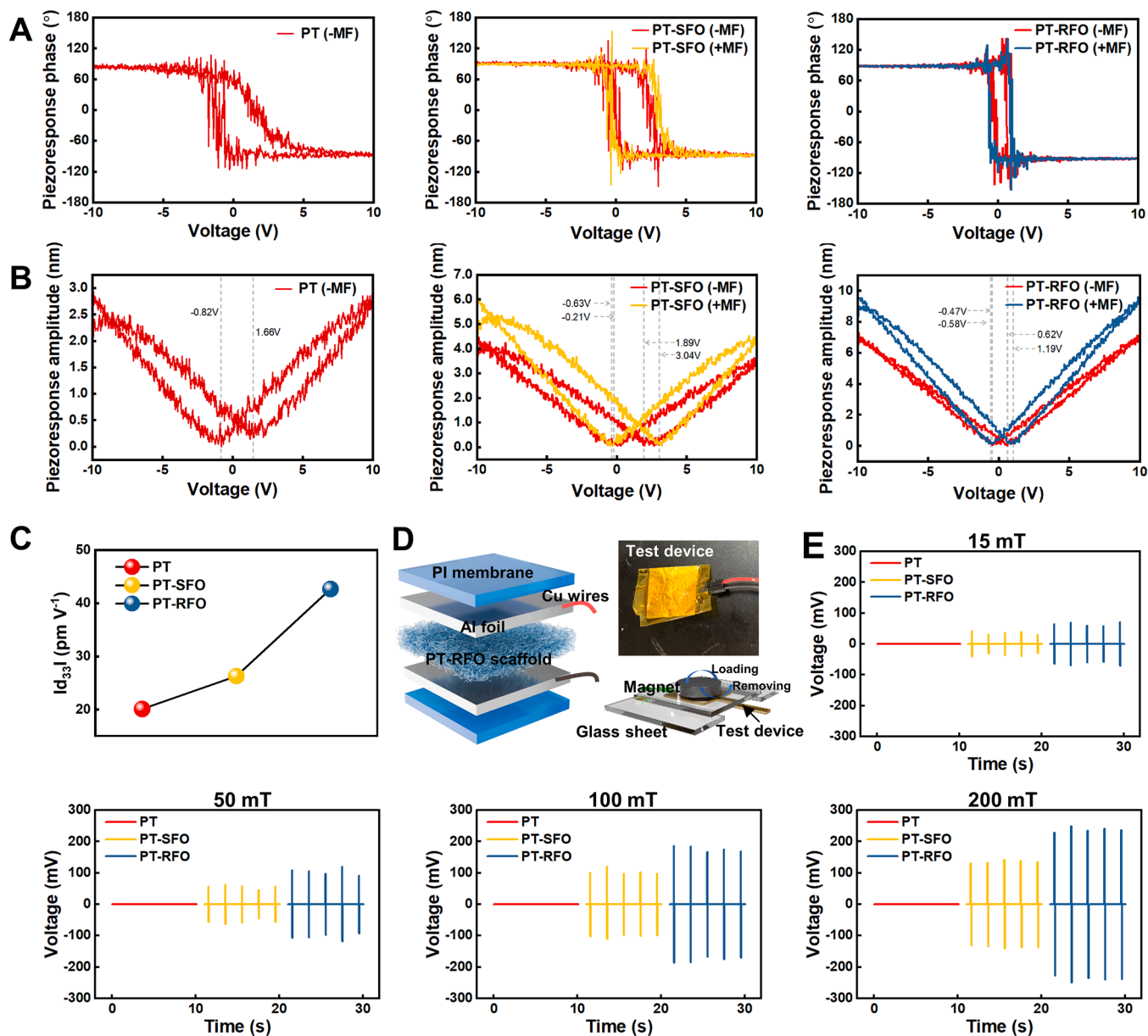


Fig. 3. Magnetic-controlled piezoelectric performance of PT-RFO scaffold. (A) Piezoresponse phase curves with and without MF (100 mT). (B) Amplitude-voltage curves with and without MF (100 mT). (C) $|d_{33}|$ values. (D) The schematic of a flexible device for measuring the output voltage of the three scaffolds. (E) The voltage of the PT-RFO scaffold under different MF intensities by periodically removing and placing a magnet.

voltage under same force. Considering the magnetic-mechano-electric cascade, the magnitude of force originated from magnetic-mechano response of magnetic NPs would determine the generated output voltage. Then, we investigated the difference in magnetic-mechano response and mechano-electric conversion of PT-SFO and PT-RFO scaffolds under MF loading. First, the strain distribution and electric potential obtained from the PT-SFO and PT-RFO scaffolds were simulated and investigated with finite element analysis. In a simplified model, two ends of the magnetic NPs, including SFO and RFO NPs, were used as the fixed constraint points within a single PVDF-TrFE nanofiber, along with the direction of a nanofiber. The strain distributions of PT-SFO and PT-RFO scaffolds applied with different MF intensities were shown in Fig. 4A & S12, and the strain at the contact interface between magnetic NPs and PVDF-TrFE nanofiber of PT-RFO scaffold reached to the value of 1.18×10^{-11} nm, which was quite higher than that of PT-SFO scaffold (3.05×10^{-12} nm) even under a weak external MF (15 mT). Due to the anisotropic morphology, RFO NPs with sharp ends generated higher

strain, compared to isotropic SFO NPs (Fig. 4B). With the increasing of applied external MF, the highest strain around the surface of SFO and RFO NPs also increased, respectively. Obviously, RFO NPs triggered higher strain under the same MF. As a result, PT-RFO group yielded piezoelectric potential going from 8.13×10^{-12} V to 1.83×10^{-9} V with external MF increased from 15 mT to 200 mT (Fig. 4C, D & S13). The theoretical piezoelectric potential trend of PT-RFO > PT-SFO was also consistent with the β -phase content and strain distribution within these two scaffolds.

Actually, unlike the simplified model, the anisotropic RFO NPs are not always aligned along the long axis of PVDF-TrFE nanofibers. As Fig. 4E schemed, for PT-RFO scaffold in a MF (along with the direction of nanofibers), besides the pressure force parallel to MF direction (F_{parallel}), the additional deflection force was generated and applied onto the contact interface of PVDF-TrFE and RFO NPs, as the anisotropic RFO NPs tended to deflect in the direction parallel to the MF. After force analysis, the deflection force can be decomposed into two forces loaded onto the

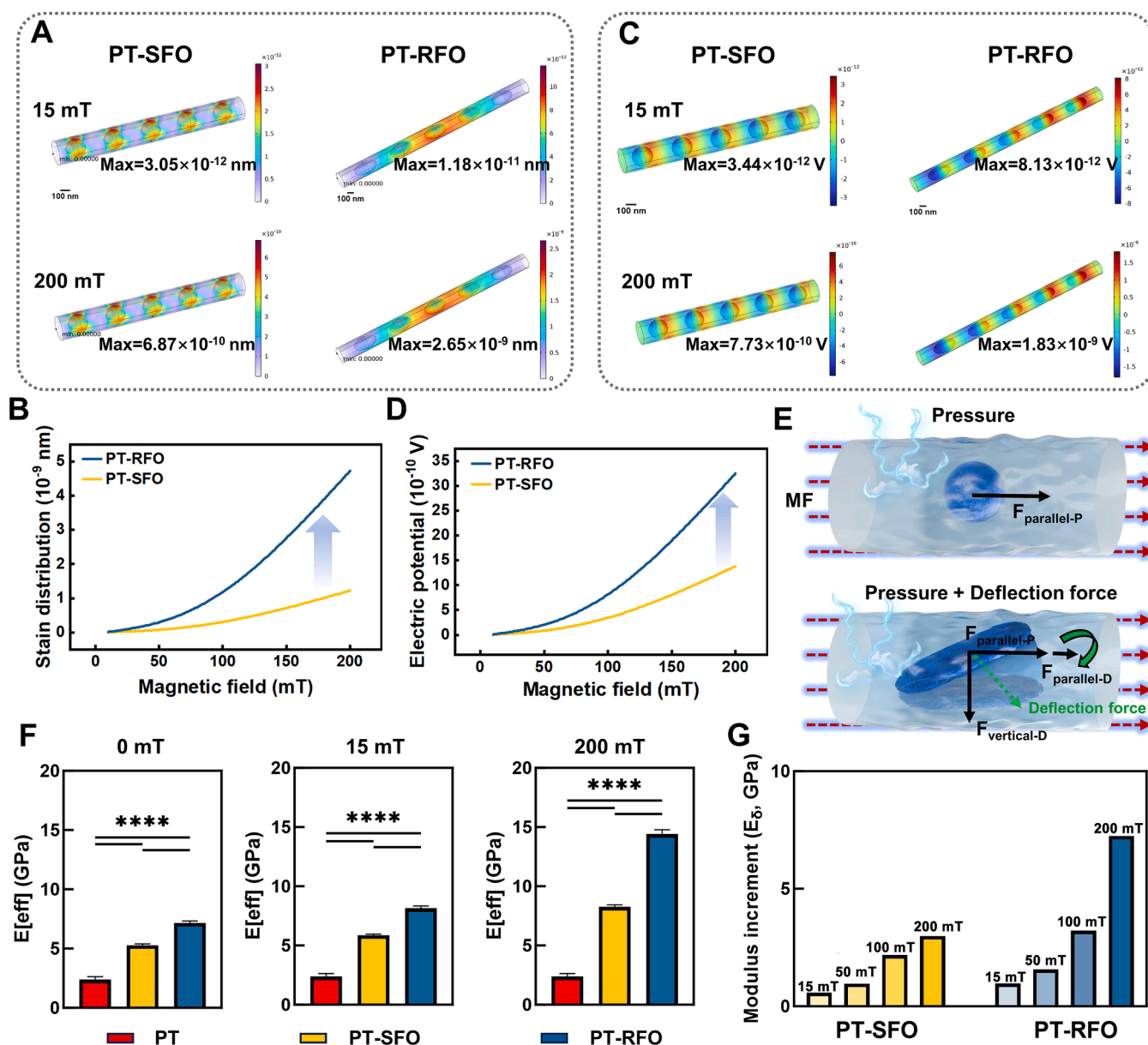


Fig. 4. Investigation of mechano-electric conversion within PT-RFO scaffold. (A) COMSOL simulation of stain distribution of PT-SFO and PT-RFO scaffolds under 15 mT and 200 mT. (B) Stain distribution-MF intensity curves. (C) COMSOL simulation of electric potential of PT-SFO and PT-RFO scaffolds under different MF intensity. (D) Electric potential-MF intensity curves. (E) Schematic of deflection force within PT-RFO scaffold. (F) Surface modulus of PT, PT-SFO and PT-RFO scaffolds under different MF intensity using nano-indenter test. (G) Modulus increment of PT-SFO and PT-RFO scaffolds under different MF intensity.

PVDF-TrFE, which are parallel ($F_{\text{parallel-D}}$) and vertical ($F_{\text{vertical-D}}$) to the MF direction, thus resulting higher piezoelectric effects. However, isotropic SFO NPs had not deflection force within PT-SFO scaffold. To verify this assumption, we adopted nano-indenter to assess the modulus changes of PT-SFO and PT-RFO scaffolds under MF loading. As shown in Fig. 4F & S14, pure PT scaffold kept stable modulus of 2.37 GPa with or without MF. It is worth noting that, even without MF, PT-SFO (5.28 GPa) and PT-RFO (7.16 GPa) scaffolds showed a little increased modulus, although we used the same concentration of NPs during their respective preparation procedures. This difference of modulus should come from the presence of harder Fe₃O₄ NPs, which were inevitably located within the micro testing area by nano-indenter, and as a matter, RFO NPs with longer major axis produced higher modulus. Put this aside, it's worth to noting the increment of modulus with the MF loading intensity increasing. Especially, under a MF, 15 mT for example, the modulus increment (E_0) of PT-RFO scaffold (0.97 GPa) was greater than that of

PT-SFO scaffold (0.58 GPa) (Fig. 4G). Simultaneously, higher MF intensity resulted in larger variation in the modulus increment within PT-SFO and PT-RFO scaffolds. That is to say, the same MF intensity can trigger higher mechano-electric conversion within PT-RFO scaffold, due to the existence of deflection force. In summary, due to the magneto-responsiveness, on one hand, the magnetic phase of the nanorods undergoes magnetostriction along the magnetic-field direction, inducing a magnetic deformation of the nanorods; this deformation is then transferred directly to the PVDF matrix, producing a direct pressure. On the other hand, the magnetic domains in nanorods whose axes are not fully aligned with the MF rotate toward that direction, and the deflection force generated during this rotation likewise acts on the PVDF matrix, which can further enhance the electrical output. When a time-varying MF is applied over a period, such as a periodically applied pulsed MF, the anisotropic magnetic nanorods repeatedly undergo these processes, including shape deformation-recovery and long-axis deflection-

recovery. As a result, the magneto-mechano-electric cascade scaffolds can continuously output electrical signals for subsequent regulation of cell behavior.

3.5. Magneto-mechano-electric cascade PT-RFO scaffold regulated cellular behaviors in vitro

Next, we tried to investigate the effect of magneto-mechano-electric cascade on cellular behaviors. Considering the basic role of cell adhesion on supporting cell function, which is highly dependent on the cell adhesion domains in matrix (e.g. RGD) [45], we uniformly coated collagen onto three scaffolds by electrostatic spraying before cell seeding (Figure S15). Though the piezoelectric property was slightly affected, the $|d_{33}|$ value of PT-RFO scaffold (40.76 pm/V) is still higher than most PVDF-based materials in Table S2 (Figure S16 & S17). For the sake of simplicity, the PT, PT-SFO and PT-RFO groups represented the corresponding scaffolds with collagen coating in following experiments.

In clinical practice, transcranial magnetic stimulation uses the MF stimulation parameter of exceeding 1 T, but it is usually accompanied by adverse reactions like scalp pain, neck soreness, a burning sensation, and, in severe cases, seizure induction [46]. Hence, recent researches focus more on the parameters of low-intensity MF stimulation. Neuronal excitation needs 15–30 mV membrane depolarization, but a part of the

applied potential will dissipate across the interface between extracellular electrical stimulation and cell [47,48]. Considering the gentler stimulation mode and effective membrane depolarization, the MF intensity of 15 mT is chosen for the subsequent experiments, which is much mild to output high-efficiency electrical voltage compared to the reported studies (Table S3) [15,18,19].

We first investigated the biocompatibility of three scaffolds. PT, PT-SFO and PT-RFO scaffolds were co-cultured with PC12 cells with MF loading. The PC12 cell line model is widely used in neurobiology studies because of its neurogenic characteristics upon differentiation, such as neuro-related bioactive factor secretion and neuron morphology [49–51]. Figure S18 indicates that all scaffolds showed a progressive increased PC12 cells in each group as prolonging culture time, exhibiting good biocompatibility. Ulteriorly, we assessed the influence of PT, PT-SFO and PT-RFO scaffolds coupled with a pulsed MF on cell spreading. The cell morphology demonstrated that PC12 cells tightly adhered onto the three scaffolds with obvious cell spreading (Fig. 5A & S19). By statistical analysis, the PT-RFO (+MF) group showed the highest cell length. On one hand, the magnetic NPs encapsulated in PT-SFO and PT-RFO scaffolds produced magneto-mechanic effect by mediating immediate stiffening under MF loading (Fig. 4F), which in turn promoted mechanical-sensitive cytoskeleton assembly [52]. On the other hand, the mechano-electric effect outputted ES, which then activated

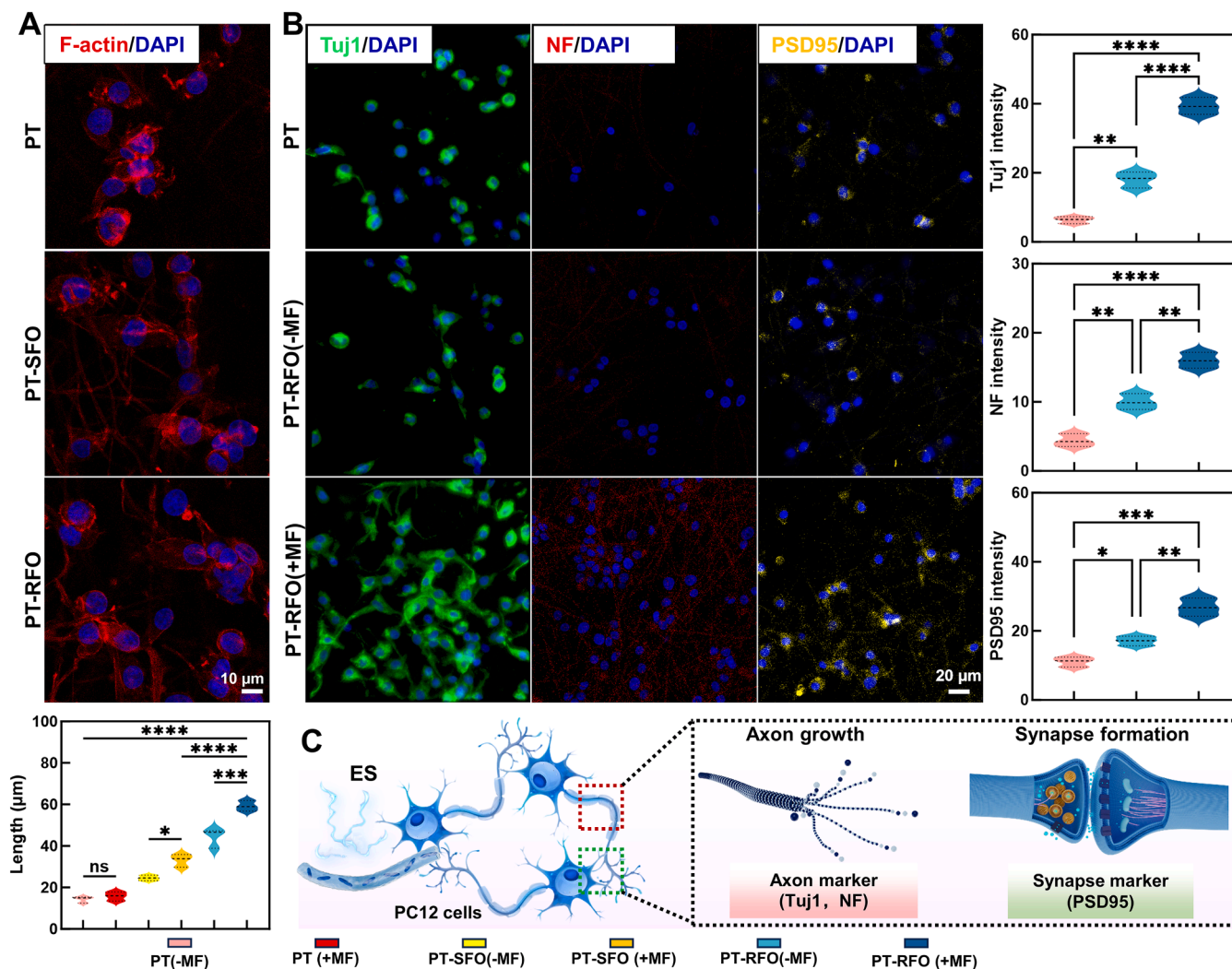


Fig. 5. Magneto-mechano-electric cascade PT-RFO scaffold promoted PC12 cells spreading and neural functional expression. (A) F-actin staining images of PC12 cells cultured on PT, PT-SFO and PT-RFO scaffolds under MF loading. (B) Representative immunofluorescence images of Tuj-1 (green), NF (red) and PSD95 (yellow) of PC12 cells on three scaffolds under MF loading. (C) Schematic of the PT-RFO scaffold promotes axon regeneration and synaptogenesis of PC12 cells.

intracellular signaling pathways affecting the dynamic balance of the cytoskeleton. This is consistent with electrical stimulated cell differentiation demonstrated by other piezoelectric materials studies [53,54].

Then, we investigated the influence of magneto-mechano-electric cascade on the neural functional expression of PC12 cells under MF loading for 7 days. The neuron-specific microtubule element marker β -Tubulin (Tuj1), neurofilament (NF) and post-synaptic density protein 95 (PSD95) staining were performed to study the neural functional expression of PC12 cells. As shown in the immunofluorescence images (Fig. 5B), an obvious increasing trend of PT < PT-RFO(-MF) < PT-RFO(+MF) was observed in Tuj1, NF and PSD95 expression. Usually, Tuj1 and NF are respectively identify regenerative axons, and PSD95 is respectively evaluate synaptogenesis [55]. The fluorescence intensity has been widely adopted to indicate cellular responses [56]. The intensity of Tuj1, NF and PSD95 makers showed that the production of neural functional factors was enhanced in magneto-mechano-electric

cascade PT-RFO (+MF) scaffold. Thus, we can conclude that the PT-RFO scaffold with high magneto-mechano-electric conversion efficiency promoted higher axon regeneration and synaptogenesis of PC12 cells (Fig. 5C).

3.6. Magneto-mechano-electric cascade PT-RFO scaffold promoted sciatic nerve structural reconstruction and inflammatory regulation

The above results demonstrated the good ES outputs of PT-RFO scaffold coupled with MF loading for promoting neural functional expression of nerve cells *in vitro*, rendering it suitable for subsequent PNI repair therapy *in vivo*. For this purpose, the PT-RFO scaffold was warped onto the sciatic nerve in Sprague-Dawley (SD) rat with a 5 mm long crushed defect (Figure S20). All rats were divided into five groups: (1) the Sham group, the rats without any treatment; (2) the PNI group, the rats with sciatic nerve injury; (3), the PT group, the PNI rats treated with

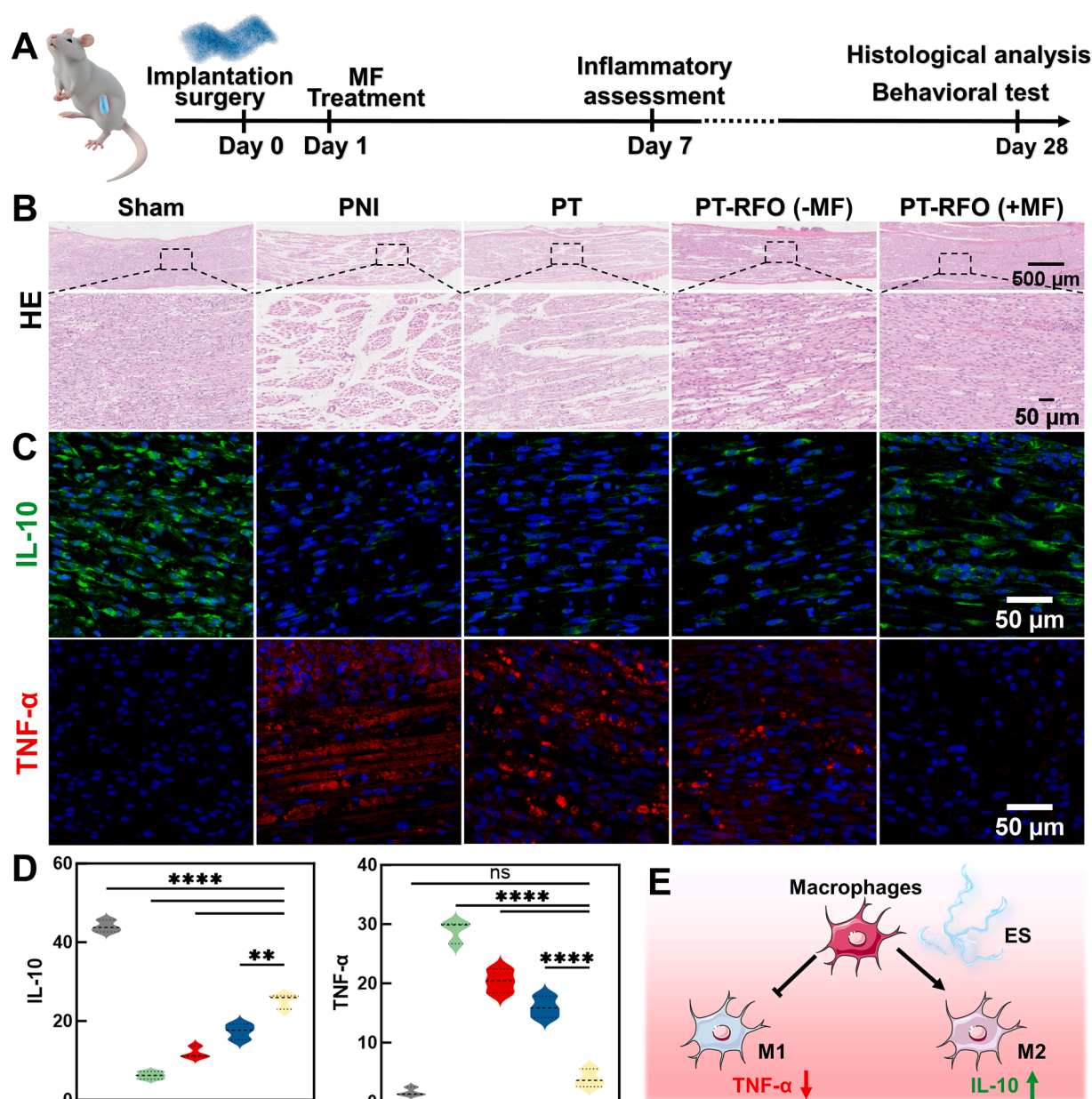


Fig. 6. (A) Schematic of treatment and repair assessment of injured sciatic nerve in rat model. (B) H&E staining images. (C) Anti-inflammatory (IL-10, green) and inflammatory (TNF- α , red) makers of macrophages at day 7. (D) Quantification of the intensities of IL-10 and TNF- α with Image J. (E) Schematic of ES promoting M2 polarization of macrophages.

PT scaffold; (4) the PT-SFO group, the PNI rats treated with PT-RFO scaffold; and (5) the PT-RFO (+MF) group, the PNI rats treated with PT-RFO scaffold under MF loading using a circular Helmholtz coil, which provides more even and deeper MF stimulation.

To investigate the histological morphology of sciatic nerves recovered after injury, the segments of sciatic nerves containing the defect site of the rats were sampled and sectioned longitudinally for the subsequent staining after 4 weeks of implantation (Fig. 6A). Hematoxylin and eosin (HE) staining results reveal that the axonal fibers in PNI group showed a sparse, hollow, and disconnected state, while the axonal fibers in PT-RFO (+MF) group showed a dense, well-aligned and connected state with similar histological morphology to the Sham group (Fig. 6B). Without MF, PT-RFO group showed an insufficient PNI repair effect, but still better than PT group. The possible reason is attributed to that PT-RFO group with anisotropic RFO NPs has more piezoelectric β -phase, which can promote the production of electrical outputs mediated by cell force during cell motility.

To assess inflammatory reaction after PNI injury, the sciatic nerve sections were immunofluorescence stained and quantitative analyzed with inflammatory maker (TNF- α) and anti-inflammatory maker (IL-10) of macrophages (Fig. 6C & D). The PNI injury significantly induced the expression of TNF- α in the PNI group after 1 weeks, while PT, PT-RFO and PT-RFO (+MF) groups reduced the expression of TNF- α , and increased the expression of IL-10. Especially, PT-RFO (+MF) group

reduced TNF- α to the level of the Sham group, and promoted the obvious expression of IL-10, indicating magneto-mechano-electric scaffolds with enough ES outputs assisted by MF loading can reduce inflammatory reaction, by promoting M2 polarization and suppressing M1 polarization of macrophages (Fig. 6E). In brief, the magneto-mechano-electric cascade PT-RFO scaffold with high ES outputs accelerated the structural reconstruction of injured sciatic nerves, and built an anti-inflammatory microenvironment, which may promote the nerve repair and neurological functional recovery.

3.7. Magneto-mechano-electric cascade scaffold promoted sciatic nerve repair and neurological functional recovery

We further investigated the effect of PT-RFO scaffold with MF loading on sciatic nerve repair and neurological functional recovery. As shown in Fig. 7A & B, the trend of Tuj-1 expression of PNI group < PT group < PT-RFO (-MF) < PT-RFO (+MF) was observed. For another indicator of budding axons, NF showed a similar tendency, displaying that the PT-RFO (+MF) possessing the highest NF expression level. Moreover, the expression of myelin structural protein, calcium-binding protein (S100), in PT-RFO (+MF) group was significantly up-regulated compared to PT-RFO (-MF) group. Taken together, these results demonstrated that the magneto-mechano-electric cascade system can promote injured sciatic nerve repair, suggesting the great potential in

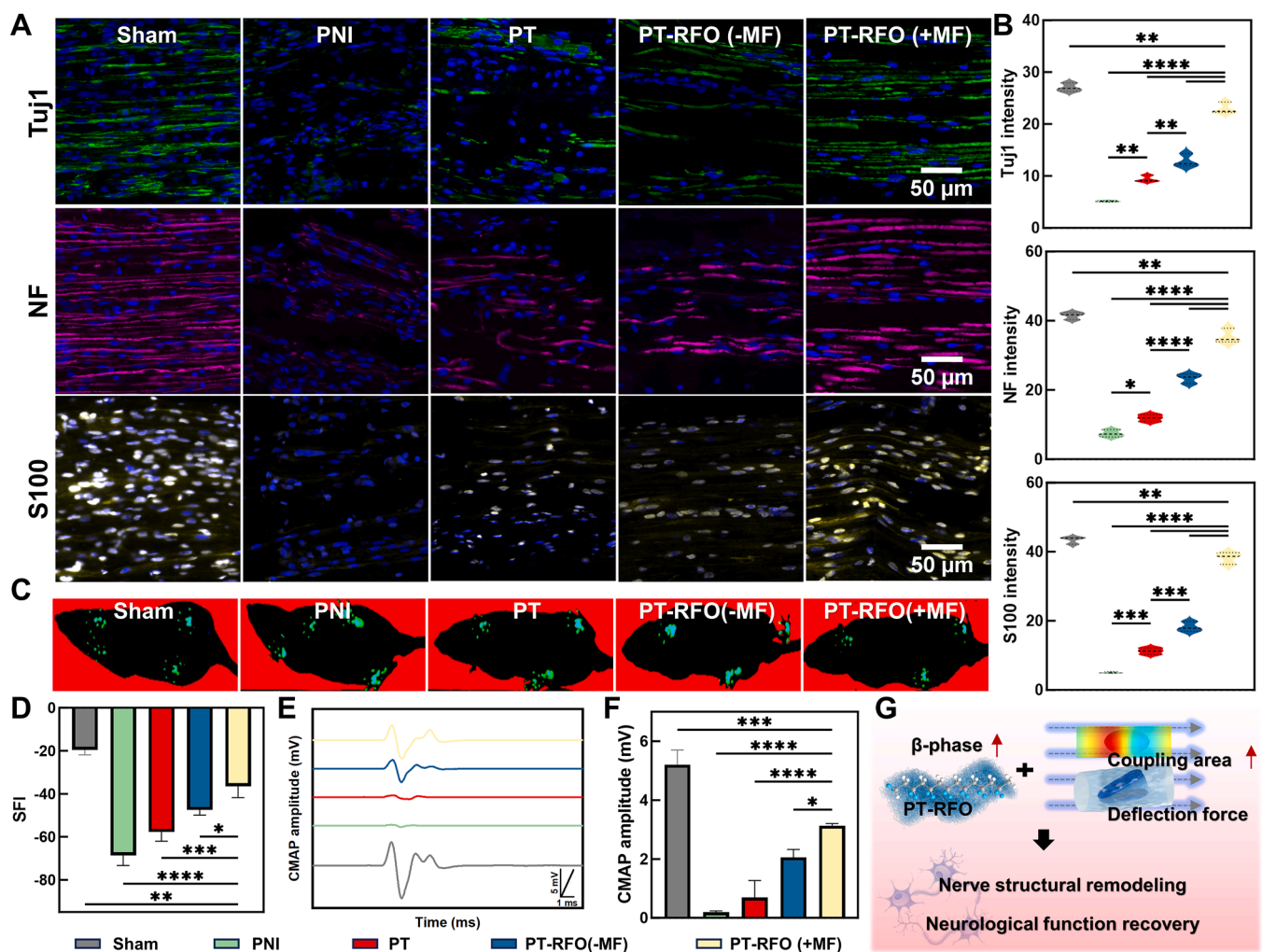


Fig. 7. Magneto-mechano-electric cascade PT-RFO scaffold with MF loading promote sciatic nerve repair and neurological functional recovery. (A) Representative immunofluorescence images and (B) quantitative intensity analysis of Tuj-1, NF-200 and S100. (C) Gait analysis. (D) SFI calculation. (E) CMAP recording curves. (F) CMAP onset-to-peak amplitude. (G) Schematic of magneto-mechano-electric cascade PT-RFO scaffold with increased β -phase content, larger coupling area and additional deflection force mediated high ES outputs for neurorestoration and neuromodulation.

nerve tissue engineering field.

Afterwards, the motor function recovery and nerve impulse conduction in rats were verified through footprint test and electrophysiological analysis, respectively. The maximum length of footprints was observed in PNI group (Fig. 7C), as lost sciatic nerves would restrict the toe extension of rats. As expected, PT-RFO (+MF) group showed the quite smaller length of footprints, which is similar with Sham group. The sciatic function index (SFI) was also used to reflect the level of sciatic nerve dysfunction, and the lower score represents the more serious injury. The SFI results showed a significantly higher recovery rate in PT-RFO (+MF) group besides from Sham group (SFI = −36.43, Fig. 7D). Moreover, we adopted electrophysiological analysis to assess the nerve conduction recovery of damaged nerves. As shown in Fig. 7E & F, the value of complex muscle action potentials (CMAP) in PT-RFO (+MF) group was significantly higher than that in PNI group, PT and PT-RFO groups, suggesting more satisfactory nerve conduction recovery. The trend was consistent with the regeneration and repair of nerve fibers in Fig. 7A, as denser nerve fibers can trigger better athletic performance. The results both proved the ability of our magneto-mechano-electric cascade PT-RFO scaffold with MF loading for the promotion of neural recovery in PNI. As summarized in Fig. 7G, the implanted magneto-mechano-electric cascade PT-RFO scaffold with higher β -phase content, larger coupling area and supplementary deflection force can trigger effective ES outputs under MF loading. The enhanced ES accelerates the nerve repair of injured sciatic nerves, and promotes neurological functional recovery of rats. This work demonstrates the potential of the magneto-mechano-electric cascade system in enhancing ES output, which can be effectively applied for electrically induced tissue repair. However, the biodegradability of PVDF-based materials is suboptimal, compared to natural polymer materials. The next step for developing electrically stimulated platforms based on cascade systems will be to explore new degradable piezoelectric materials and integrate exogenous stimuli-mediated force enhancement.

4. Conclusion

In summary, we developed a simple but effective magneto-mechano-electric cascade scaffold. By introducing anisotropic RFO NPs into PVDF-TrFE via electrospinning technique, the PT-RFO scaffold obtained high piezoelectric β -phase content. Under a remotely controllable MF loading, the pressure force and additional deflection force between RFO NPs and PVDF-TrFE matrix in PT-RFO scaffold trigger high-efficiency ES outputs. The *in vitro* studies demonstrated that PT-RFO scaffold significantly enhanced cell spreading and functional expression of PC 12 cells. In addition, *in vivo* experiments in rat models proved that our magneto-mechano-electric cascade PT-RFO scaffold accelerated the recovery of motor functions and promoted the repair of sciatic nerves. More importantly, this work offers a new cascade responsive strategy by integrating multiple physical cues in a wireless remote manner, to construct a complex microenvironment for neurorestoration and neuromodulation.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Dan Wei: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Xurui Zhao:** Methodology, Data curation. **Mingze Zeng:** Methodology, Investigation. **Rong Li:** Methodology, Investigation. **Wei Zhang:** Methodology, Investigation. **Alina Urakova:** Software, Methodology. **Jie Ding:** Methodology, Investigation, Funding acquisition. **Chengheng Wu:** Investigation, Funding acquisition. **Jing Sun:** Investigation, Funding acquisition. **Roman V. Chernozem:** Writing – review & editing, Software,

Methodology. **Hongsong Fan:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grant numbers: 32361133548 to H.F., 52373150 to J.S. and 52303195 to C.W.) and the Russian Science Foundation (№ 24-43-00171).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.actbio.2025.11.007.

References

- [1] C.D. McCaig, A.M. Rajnicek, B. Song, M. Zhao, Controlling cell behavior electrically: current views and future potential, *Physiol. Rev.* 85 (2005) 943–978.
- [2] Q. Wang, H. Wang, Y. Ma, X. Cao, H. Gao, Effects of electroactive materials on nerve cell behaviors and applications in peripheral nerve repair, *Biomater. Sci.* 10 (21) (2022) 6061–6076.
- [3] F. Jin, T. Li, T. Yuan, L. Du, C. Lai, Q. Wu, Y. Zhao, F. Sun, L. Gu, T. Wang, Physiologically self-regulated, fully implantable, battery-free system for peripheral nerve restoration, *Adv. Mater.* 33 (48) (2021) 2104175.
- [4] F. Jin, T. Li, Z. Wei, R. Xiong, L. Qian, J. Ma, T. Yuan, Q. Wu, C. Lai, X. Ma, Biofeedback electrostimulation for bionic and long-lasting neural modulation, *Nat. Commun.* 13 (1) (2022) 5302.
- [5] C. Günter, J. Delbeke, M. Ortiz-Catalan, Safety of long-term electrical peripheral nerve stimulation: review of the state of the art, *J. Neuroeng. Rehabil.* 16 (2019) 1–16.
- [6] R.e. Gilron, S. Little, R. Perrone, R. Wilt, C. de Hemptinne, M.S. Yaroshinsky, C. A. Racine, S.S. Wang, J.L. Ostrem, P.S. Larson, Long-term wireless streaming of neural recordings for circuit discovery and adaptive stimulation in individuals with Parkinson's disease, *Nat. Biotechnol.* 39 (9) (2021) 1078–1085.
- [7] A. Aabid, M.A. Raheman, Y.E. Ibrahim, A. Anjum, M. Hrairi, B. Parveez, N. Parveen, J. Mohammed Zayan, A systematic review of piezoelectric materials and energy harvesters for industrial applications, *Sensors* 21 (12) (2021) 4145.
- [8] L. Zhang, S. Li, Z. Zhu, G. Rui, B. Du, D. Chen, Y.F. Huang, L. Zhu, Recent progress on structure manipulation of poly (vinylidene fluoride)-based ferroelectric polymers for enhanced piezoelectricity and applications, *Adv. Funct. Mater.* 33 (38) (2023) 2301302.
- [9] Q. Zhao, Y. Yang, G. Xiong, J. Chen, T. Xu, Q. Xu, R. Zhang, W. Yao, H. Li, C.-S. Lee, Calcium single atom confined in nitrogen-doped carbon-coupled polyvinylidene fluoride membrane for high-performance piezocatalysis, *J. Am. Chem. Soc.* 146 (24) (2024) 16648–16658.
- [10] L. Wu, Z. Jin, Y. Liu, H. Ning, X. Liu, N.Hu Alamusi, Recent advances in the preparation of PVDF-based piezoelectric materials, *Nanotechnol. Rev.* 11 (1) (2022) 1386–1407.
- [11] H. Abdolmaleki, A.B. Haugen, K.B. Buhl, K. Daasbjerg, S. Agarwala, Interfacial engineering of PVDF-TrFE toward higher piezoelectric, ferroelectric, and dielectric performance for sensing and energy harvesting applications, *Adv. Sci.* 10 (6) (2023) 2205942.
- [12] Z. Wang, Y. Tai, Z. Ye, J. Nam, Y. Yin, Integration of ZnS: mn2⁺ microparticles into electrospun PVDF-based nanofibers for enhanced mechanoluminescence, *Adv. Funct. Mater.* 34 (51) (2024) 2410358.
- [13] D. Lei, N. Hu, L. Wu, H. Ning, Y. Wang, Z. Jin, Y. Liu, Improvement of the piezoelectricity of PVDF-HFP by CoFe2O4 nanoparticles, *Nano Mater. Sci.* 6 (2) (2024) 201–210.
- [14] J. Choi, K.T. Nam, E.-H. Sohn, Y. Seo, Macroscopic room-temperature magnetoelectricity in piezoelectric (Core)–Ferrimagnetic (Shell) nanocomposites, *ACS. Nano* 18 (44) (2024) 30681–30689.
- [15] Q. Zhang, J. Zhu, H. Liu, X. Fei, M. Zhu, Magneto-mechano-electric cascade stimulation system accelerates wound healing constructed by biodegradable magnetoelectric nanofibers, *Adv. Funct. Mater.* 34 (9) (2024) 2309968.
- [16] C. Shuai, C. Lin, C. He, W. Tan, S. Peng, W. Yang, Exchange-coupled bi-magnetic nanoparticles enhance magnetothermal/chemodynamic antibacterial therapy of poly-L-lactide scaffold, *J. Colloid. Interface Sci.* 685 (2025) 1131–1142.
- [17] C. Shuai, C. Lin, X. Gao, W. Tan, X. Bai, W. Yang, Bifunctional cobalt ferrite/reduced graphene oxide heterojunction enhances the antibacterial and osteogenic activities of scaffold, *Appl. Surf. Sci.* 695 (2025) 162942.
- [18] L. Wu, Y. Liu, H. Gao, W. Guan, S. Sun, H. Liu, X. Chu, G. Li, *In situ* non-invasive electrical stimulation by magnetically-actuated piezoelectric Fe3O4/PVDF

- alignment scaffolds for synergistically promoting long-distance peripheral nerve regeneration, *Chem. Eng. J.* 509 (2025) 161308.
- [19] Y. Yang, P. Wen, X. Chen, Y. Wang, S. Zhu, Z. Ni, L. Yuan, L. Shan, P. Zhang, P. Shi, Mechanically-compliant magnetoelectric sutures for wound management, *Adv. Funct. Mater.* (2025) 2510353.
 - [20] B. Dutta, E. Kar, N. Bose, S. Mukherjee, NiO@ SiO₂/PVDF: a flexible polymer nanocomposite for a high performance human body motion-based energy harvester and tactile e-skin mechanosensor, *ACS Sustain. Chem. Eng.* 6 (8) (2018) 10505–10516.
 - [21] Q. Zhang, J. Zhu, H. Liu, X. Fei, M.J.A.F.M. Zhu, Magneto-mechano-electric cascade stimulation system accelerates wound healing constructed by biodegradable magnetoelectric nanofibers, 34(9) (2024) 2309968.
 - [22] Y. Su, C. Dagdeviren, R.J.A.F.M. Li, Measured output voltages of piezoelectric devices depend on the resistance of voltmeter, 25(33) (2015) 5320–5325.
 - [23] A. Urakova, A. Baksheev, A. Pryadko, I. Grubova, M. Surmeneva, P. Chernozem, Y. Mukhortova, D. Wagner, E. Gerasimov, S. Kazantsev, Sub-20-nm magnetite-based core-shell nanoparticles with strong magnetic, magnetoelectric, and nanocatalytic properties, *Ceram. Int.* (2025).
 - [24] X. Guan, S. Yan, Q. Zeng, Z. Xu, Y. Chen, H. Fan, Polyacrylic acid-grafted magnetite nanoparticles for remediation of Pb (II)-contaminated water, *Fibers Polym.* 17 (2016) 1131–1139.
 - [25] Y. Xing, Q. Gao, Y. Zhang, L. Ma, K.Y. Loh, M. Peng, C. Chen, Y. Cui, The improved sensitive detection of C-reactive protein based on the chemiluminescence immunoassay by employing monodispersed PAA-Au/Fe₃O₄ nanoparticles and zwitterionic glycerophosphoryl choline, *J. Mater. Chem. B* 5 (21) (2017) 3919–3926.
 - [26] A.S. Pryadko, A.Y.R. Mukhortova, R.V. Chernozem, I. Pariy, S.I. Alipkina, I. I. Zharkova, A.A. Dudun, V.A. Zhuikov, A.M. Moisenovich, G.A. Bonartseva, Electrospun magnetic composite poly-3-hydroxybutyrate/magnetite scaffolds for biomedical applications: composition, structure, magnetic properties, and biological performance, *ACS. Appl. Bio Mater.* 5 (8) (2022) 3999–4019.
 - [27] T. Tansel, Effect of electric field assisted crystallisation of PVDF-TrFE and their functional properties, *Sens. Actuators. A Phys.* 332 (2021) 113059.
 - [28] K. Gao, X. Hu, C. Dai, T. Yi, Crystal structures of electrospun PVDF membranes and its separator application for rechargeable lithium metal cells, *Mater. Sci. Eng. B* 131 (1–3) (2006) 100–105.
 - [29] T. Lei, L. Yu, G. Zheng, L. Wang, D. Wu, D. Sun, Electrospinning-induced preferred dipole orientation in PVDF fibers, *J. Mater. Sci.* 50 (12) (2015) 4342–4347.
 - [30] V. Botvin, A. Fetisova, Y. Mukhortova, D. Wagner, S. Kazantsev, M. Surmeneva, A. Kholkin, R. Surmenev, Effect of Fe₃O₄ nanoparticles modified by citric and oleic acids on the physicochemical and magnetic properties of hybrid electrospun P (VDF-TrFE) scaffolds, *Polymers. (Basel)* 15 (14) (2023) 3135.
 - [31] R. Gonçalves, P. Martins, C. Caparrós, P. Martins, M. Benelmekki, G. Botelho, S. Lanceros-Méndez, A. Lasheras, J. Gutiérrez, J. Barandiarán, Nucleation of the electroactive β -phase, dielectric and magnetic response of poly (vinylidene fluoride) composites with Fe₂O₃ nanoparticles, *J. Non-Cryst. Solids* 361 (2013) 93–99.
 - [32] P. Martins, C.M. Costa, M. Benelmekki, G. Botelho, S. Lanceros-Mendez, On the origin of the electroactive poly (vinylidene fluoride) β -phase nucleation by ferrite nanoparticles via surface electrostatic interactions, *CrystEngComm.* 14 (8) (2012) 2807–2811.
 - [33] M. Sebastian, A. Larrea, R. Gonçalves, T. Alejo, J. Vilas, V. Sebastian, P. Martins, S. Lanceros-Méndez, Understanding nucleation of the electroactive β -phase of poly (vinylidene fluoride) by nanostructures, *RSC. Adv.* 6 (114) (2016) 113007–113015.
 - [34] M. Sahihi, A. Jaramillo-Botero, W.A. Goddard III, F. Bedoui, Interfacial interactions in a model composite material: insights into $\alpha \rightarrow \beta$ phase transition of the magnetite reinforced poly (Vinylidene Fluoride) systems by all-atom molecular dynamics simulation, *J. Phys. Chem. C* 125 (39) (2021) 21635–21644.
 - [35] F.S.N. Oliva, M. Sahihi, L. Lenglet, A. Ospina, E. Guenin, A. Jaramillo-Botero, W. A. Goddard III, F. Bedoui, Nanoparticle size and surface chemistry effects on mechanical and physical properties of nano-reinforced polymers: the case of PVDF-Fe₃O₄ nano-composites, *Polym. Test.* 117 (2023) 107851.
 - [36] D. Yuan, Z. Li, W. Thitsartarn, X. Fan, J. Sun, H. Li, C. He, β phase PVDF-hfp induced by mesoporous SiO₂ nanorods: synthesis and formation mechanism, *J. Mater. Chem. C* 3 (15) (2015) 3708–3713.
 - [37] M. Baniasadi, Z. Xu, S. Hong, M. Naraghi, M. Minary-Jolandan, Thermo-electromechanical behavior of piezoelectric nanofibers, *ACS Appl. Mater.* 8 (4) (2016) 2540–2551.
 - [38] Y. Zhang, X. Wu, J. Ding, B. Su, Z. Chen, Z. Xiao, C. Wu, D. Wei, J. Sun, F. Luo, Wireless-powering deep brain stimulation platform based on 1D-structured magnetoelectric nanochains applied in antiepilepsy treatment, *ACS. Nano* 17 (16) (2023) 15796–15809.
 - [39] T. Roopaa, H.N. Murthy, V.P. Kumar, M. Krishna, Development and characterization of PVDF thin films for pressure sensors, *Mater. Today* 5 (10) (2018) 21082–21090.
 - [40] C.M. Wu, M.H. Chou, Polymorphism, piezoelectricity and sound absorption of electrospun PVDF membranes with and without carbon nanotubes, *Compos. Sci. Technol.* 127 (2016) 127–133.
 - [41] H.-J. Tseng, W.-C. Tian, W.-J. Wu, P (VDF-TrFE) polymer-based thin films deposited on stainless steel substrates treated using water dissociation for flexible tactile sensor development, *Sensors* 13 (11) (2013) 14777–14796.
 - [42] Y. Jang, H.S. Kim, Enhanced piezoelectric nanogenerator made of electrospun PVDF and P (VDF-TrFE) nanowires with the addition of TrFE comonomers, *J. Mater. Sci. - Mater. Electron.* 34 (22) (2023) 1633.
 - [43] E. Stojchevska, P. Makreski, M. Zanon, L. Gasperini, G. Selli, D. Fabiani, C. Gualandi, A. Bužarovska, Piezoelectric PVDF-TrFE nanocomposite mats filled with BaTiO₃ nanofibers: the effect of poling conditions, *Polym. Adv. Technol.* 35 (2) (2024) e6333.
 - [44] P. Martins, Y.V. Kolen'ko, J. Rivas, S. Lanceros-Mendez, Tailored magnetic and magnetoelectric responses of polymer-based composites, *ACS Appl. Mater.* 7 (27) (2015) 15017–15022.
 - [45] A. Lagunas, I. Tsintzou, Y. Vida, D. Collado, E. Pérez-Inestrosa, C.R. Pereira, J. Magalhães, J.A. Andrades, J. Samitier, Tailoring RGD local surface density at the nanoscale toward adult stem cell chondrogenic commitment, *Nano Res.* 10 (2017) 1959–1971.
 - [46] F. Fregni, A. Pascual-Leone, Technology insight: noninvasive brain stimulation in neurology—Perspectives on the therapeutic potential of rTMS and tDCS, *Nat. Clin. Pract. Neurol.* 3 (7) (2007) 383–393.
 - [47] Y.-J. Kim, N. Kent, E. Vargas Paniagua, N. Driscoll, A. Tabet, F. Koehler, E. Malkin, E. Frey, M. Manthey, Sahasrabudhe, magnetoelectric nanodiscs enable wireless transgene-free neuromodulation, *Nat. Nanotechnol.* 20 (1) (2025) 121–131.
 - [48] I. Schoen, P. Fromherz, The mechanism of extracellular stimulation of nerve cells on an electrolyte-oxide-semiconductor capacitor, *Biophys. J.* 92 (3) (2007) 1096–1111.
 - [49] W. Pi, F. Rao, J. Cao, M. Zhang, T. Chang, Y. Han, Y. Zheng, S. Liu, Q. Li, X. Sun, Sono-electro-mechanical therapy for peripheral nerve regeneration through piezoelectric nanotracts, *Nano Today* 50 (2023) 101860.
 - [50] F. Jiang, Y. Shan, J. Tian, L. Xu, C. Li, F. Yu, X. Cui, C. Wang, Z. Li, K. Ren, Poly (l-lactic acid) nanofiber-based multilayer film for the electrical stimulation of nerve cells, *Adv. Mater. Interfaces.* 10 (17) (2023) 2202474.
 - [51] A. Rajan Unnithan, V. Krishnamoorthi Kaliannagounder, N. Rao Alluri, C.H. Park, P. Veluswamy, A.J.A.N.R. Ramachandra Kurup Sasikala, Design and application of piezoelectric conductive smart scaffold for noninvasive neural tissue regeneration via custom-made in vitro mechano-stimulator, *Adv. Nanobiomed. Res.* (2025) 2500058.
 - [52] D. Wei, M. Zeng, B. Su, Y. Zhang, J. Ding, C. Wu, J. Sun, L. Zhou, H. Yin, H. Fan, Magnetic on-off manipulated matrix mechanical vibration to enhance cell clutches-reinforcement and Ca²⁺ influx facilitating BMSCs neural differentiation and TBI repair, *Chem. Eng. J.* 484 (2024) 149521.
 - [53] T. Li, M. Qu, C. Carlos, L. Gu, F. Jin, T. Yuan, X. Wu, J. Xiao, T. Wang, W. Dong, High-performance poly (vinylidene difluoride)/dopamine core/shell piezoelectric nanofiber and its application for biomedical sensors, *Adv. Mater.* 33 (3) (2021) 2006093.
 - [54] L. Yan, C. Zhao, Y. Wang, Q. Qin, Z. Liu, Y. Hu, Z. Xu, K. Wang, X. Jiang, L. Han, Adhesive and conductive hydrogel-based therapy simultaneously targeting neuroinflammation and neurofunctional damage after brain injury, *Nano Today* 51 (2023) 101934.
 - [55] X. Liu, C. Wu, Y. Zhang, G. Li, S. Chen, Z. Chen, P. Liu, K. Wu, X. Wu, T. Zhou, Viscoelastic cues to induce stem cell migration and neuronal differentiation in cell-free hydrogel-assisted TBI recovery, *Chem. Eng. J.* 492 (2024) 152401.
 - [56] S. Wang, H. Lu, X. Kang, Z. Wang, S. Yan, X. Zhang, X. Shi, Electroconductive and immunomodulatory natural polymer-based hydrogel bandages designed for peripheral nerve regeneration, *Adv. Funct. Mater.* 34 (8) (2024) 2310903.